Zero Hunger (SDG 2) Time-series analysis for predicting future trends in food prices, enabling better planning for food security

AN APPLICATION OF TIME SERIES ANALYSIS FOR FOOD PRICES IN THE PHILIPPINES

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Introduction

Importance of Price Forecasting

- Crucial across various industries (e.g., finance, agriculture, manufacturing) for informed decision-making.
- Accurate forecasting aids in profitability, operational efficiency, and risk management.

Time Series Forecasting and ARIMA

- Uses historical data to predict future trends, focusing on identifying patterns.
- ARIMA (Autoregressive Integrated Moving Average) is widely used for its ability to model trends and seasonality in univariate time series data.
- Known challenges include parameter selection and managing data variability.

Introduction

Objective of the Study

- Utilize the Orange data mining platform for ARIMA implementation, making it more accessible for non-technical users.
- Assess ARIMA's effectiveness in forecasting price trends, particularly in the context of food prices in the Philippines.

Research Contribution

- Provides insights into ARIMA's usability for price forecasting.
- Validates Orange as a tool for simplifying complex forecasting processes.

Background and Related Work

Stock Price Forecasting

- Adebiyi, Adewumi, and Ayo: Developed an ARIMA model tailored for stock price prediction using historical data, focusing on model selection based on Bayesian Information Criterion (BIC) and adjusted R-squared for accuracy.

Systematic Stock Price Forecasting

- Mondal, St, and Goswami: Used a structured approach to ARIMA model selection for stock price forecasting, focusing on optimizing parameters (p, d, q) based on the Akaike Information Criterion (AIC and AICc).

Background and Related Work

Agricultural Price Forecasting

- Jadhav, Reddy, and Gaddi: Applied ARIMA for predicting prices of cereal crops (e.g., paddy, ragi, maize) in Karnataka, incorporating seasonal indices to account for cyclical price changes in agricultural markets.

Electricity Price Forecasting

- Contreras, Espínola, Nogales, and Conejo: Employed ARIMA in competitive electricity markets, with adaptations for multi-seasonal differencing to handle daily and weekly price variations.
- Conejo, Plazas, Espínola, and Molina: Combined ARIMA with wavelet transformation to improve accuracy, decomposing price series into stable components before forecasting.

Data Collection

Data Source

- World Food Programme (WFP) Price Database.
- Covers staple food items: maize, rice, beans, fish, and sugar across various Philippine markets.

Dataset Details

- Monthly data from 2019 onward, ensuring relevance to current trends.
- Represents diverse regions, allowing analysis of regional price fluctuations and seasonal trends.

Purpose

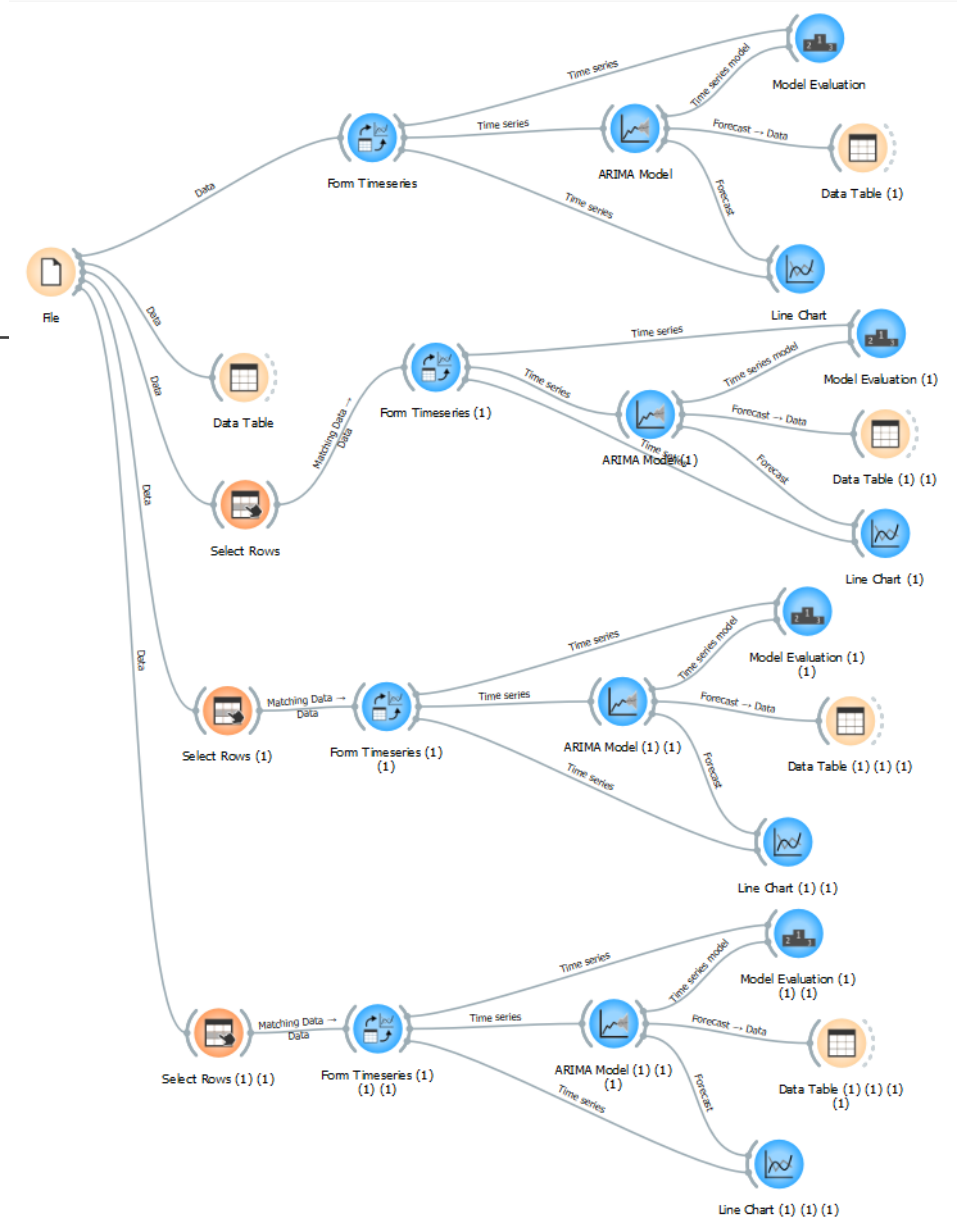
- Provides a robust foundation for ARIMA-based forecasting.
- Enables insights into price dynamics essential for stakeholders in the food supply chain.

Data Collection

price	date	admin1	admin2	market	latitude	longitude	category	commodity	unit	priceflag	pricetype	currency	usdprice
65.85	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	cereals and tub...	Potatoes (Irish)	KG	actual	Retail	PHP	1.2663
40.60	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	cereals and tub...	Rice (regular, m...	KG	actual	Retail	PHP	0.7808
37.46	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	cereals and tub...	Sweet potatoes	KG	actual	Retail	PHP	0.7204
180.31	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	meat, fish and ...	Fish (fresh)	KG	actual	Retail	PHP	3.4675
148.00	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	meat, fish and ...	Meat (chicken, ...	KG	actual	Retail	PHP	2.8462
220.31	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	meat, fish and ...	Meat (pork)	KG	actual	Retail	PHP	4.2367
30.51	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	vegetables and ...	Cabbage	KG	actual	Retail	PHP	0.5867
25.32	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	vegetables and ...	Cabbage	KG	actual	Wholesale	PHP	0.4869
46.00	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	vegetables and ...	Carrots	KG	actual	Retail	PHP	0.8846
36.00	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	vegetables and ...	Carrots	KG	actual	Wholesale	PHP	0.6923
84.31	2019-01-15	Autonomous re...	Maguindanao	Shariff Aguak	6.86472	124.442	vegetables and ...	Onions (red)	KG	actual	Retail	PHP	1.6213
63.62	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	cereals and tub...	Sweet potatoes	KG	actual	Retail	PHP	1.2235
157.23	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	meat, fish and ...	Fish (fresh)	KG	actual	Retail	PHP	3.0237
300.00	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	meat, fish and ...	Meat (beef, cho...	KG	actual	Retail	PHP	5.7692
186.15	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	meat, fish and ...	Meat (chicken, ...	KG	actual	Retail	PHP	3.5798
226.00	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	meat, fish and ...	Meat (pork)	KG	actual	Retail	PHP	4.3462
55.08	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	vegetables and ...	Cabbage	KG	actual	Retail	PHP	1.0592
25.74	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	vegetables and ...	Cabbage	KG	actual	Wholesale	PHP	0.4950
63.85	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	vegetables and ...	Carrots	KG	actual	Retail	PHP	1.2279
213.85	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	vegetables and ...	Garlic	KG	actual	Retail	PHP	4.1125
102.46	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	vegetables and ...	Onions (red)	KG	actual	Retail	PHP	1.9704
92.15	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	vegetables and ...	Onions (white)	KG	actual	Retail	PHP	1.7721
21.93	2019-01-15	Cordillera Admi...	Benguet	La Trinidad	16.455	120.588	vegetables and ...	Tomatoes	KG	actual	Wholesale	PHP	0.4217
34.89	2019-01-15	National Capita...	Metropolitan M...	Metro Manila	14.6042	120.982	cereals and tub...	Maize flour (yel...	KG	actual	Retail	PHP	0.6710
44.34	2019-01-15	National Capita...	Metropolitan M...	Metro Manila	14.6042	120.982	cereals and tub...	Rice (milled, su...	KG	actual	Retail	PHP	0.8527

Experimental Setup

- **Data Acquisition and Preparation**: Data from the WFP Price Database was imported and visualized in Orange for initial inspection.
- **Data Selection and Cleaning**: Relevant data was filtered by region and product, enhancing accuracy by isolating instances and removing outliers.
- **Time Series Transformation**: Data was converted into time series format, structuring it for ARIMA analysis.
- **Model Implementation**: The ARIMA model was applied with chosen parameters to generate forecasts with confidence intervals.
- **Forecast Visualization**: Forecasted and actual data were visualized using customizable line charts.
- **Model Evaluation**: Model accuracy was assessed using RMSE, MAE, MAPE, and R^2 metrics with cross-validation.
- **Forecasting Process Summary**: The structured process ensured thorough data filtering, transformation, forecasting, and result visualization for trend analysis.



Forecasting Process and Evaluation

- Forecasting Process: Data was imported, filtered, transformed into time series, and fitted to the ARIMA model, with results visualized to show trends and uncertainties.
- Model Evaluation: The ARIMA model's performance was evaluated using error metrics (RMSE, MAE, MAPE, R^2) and cross-validation to ensure accuracy and stability.

Results and Discussion

Data Table Overview

- Displayed forecasted prices with corresponding 95% confidence intervals, showing an upward trend in food prices.

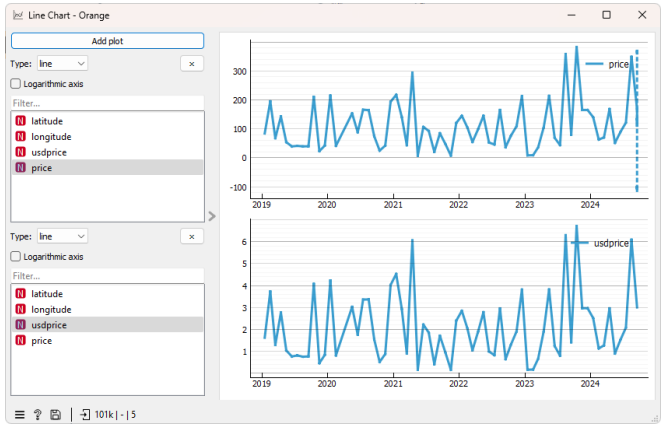
Line Charts

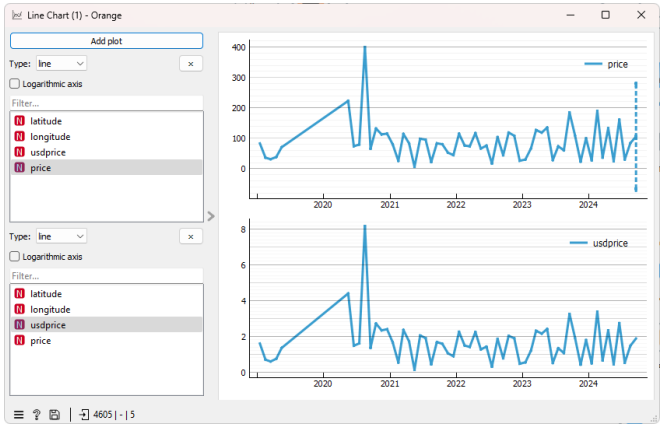
- Visualized actual vs. forecasted price trends across regions, providing a clear comparison and highlighting regional variations.

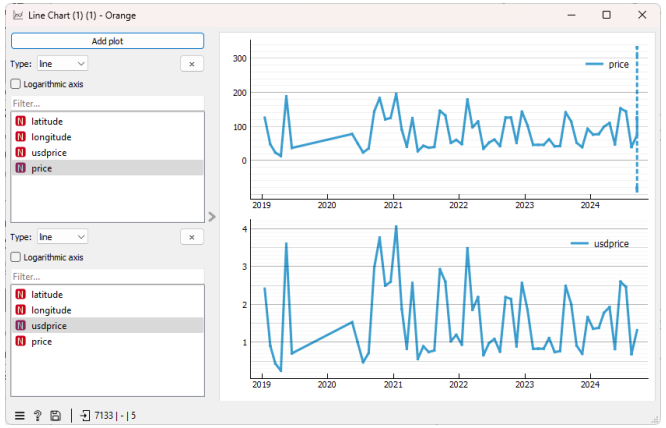
Model Evaluation Metrics

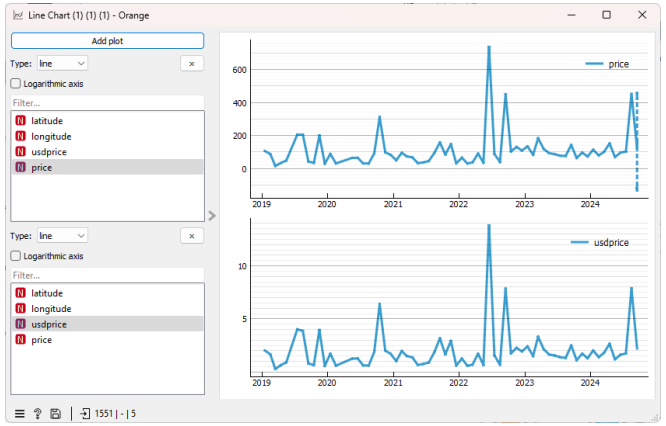
- Metrics (RMSE, MAE, MAPE) confirmed high accuracy, with cross-validation supporting model stability.

Results and Discussion









Results and Discussion

Data Table (1) - Orange

Info
5 instances (no missing data)
3 features
No target variable.
No meta attributes.

Variables
Show variable labels (if present)
Visualize numeric values
Color by instance classes

Selection
Select full rows

Restore Original Order
Send Automatically

	price (forecast)	price (95%CI low)	price (95%CI high)
1	110.864	-114.516	336.243
2	127.428	-106.853	361.708
3	132.129	-102.855	367.112
4	133.463	-101.577	368.502
5	133.841	-101.203	368.885

Data Table (1) (1) - Orange

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Send Automatically

	price (forecast)	price (95%CI low)	price (95%CI high)
1	99.0005	-72.9018	270.903
2	106.429	-67.0343	279.892
3	107.432	-66.0594	280.923
4	107.567	-65.9244	281.059
5	107.586	-65.9062	281.078

Data Table (1) (1) (1) - Orange

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	price (forecast)	price (95%CI low)	price (95%CI high)
1	108.465	-90.9869	307.916
2	121.536	-84.6945	327.766
3	124.972	-81.7183	331.662
4	125.876	-80.8465	332.598
5	126.113	-80.6112	332.837

Data Table (1) (1) (1) (1) - Orange

Info
5 instances (no missing data)
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No target variable.
No meta attributes.

Variables
Show variable labels (if present)
Visualize numeric values
Color by instance classes

Selection
Select full rows

Restore Original Order
Send Automatically

	price (forecast)	price (95%CI low)	price (95%CI high)
1	145.987	-133.262	425.236
2	164.208	-122.147	450.562
3	168.344	-118.372	455.06
4	169.283	-117.452	456.018
5	169.496	-117.239	456.232

Results and Discussion

Model Evaluation - Orange

Evaluation Parameters

Number of folds: 10

Forecast steps: 5

Apply

101k

	RMSE	MAE	MAPE	POCID	R^2	AIC	BIC
ARMA(1,0,0)	141.5	85.1	0.605	55.1	-0.054	1.249e+06	1.249e+06
ARMA(1,0,0) (in-sample)	124.0	73.5	0.686	49.5	-0.071	1.249e+06	1.249e+06

Model Evaluation (1) - Orange

Evaluation Parameters

Number of folds: 10

Forecast steps: 5

Apply

4605

	RMSE	MAE	MAPE	POCID	R^2	AIC	BIC
ARMA(1,0,0)	76.6	52.6	0.619	55.1	0.003	53121	53140
ARMA(1,0,0) (in-sample)	83.7	51.7	0.565	44.0	-0.008	53722	53741

Model Evaluation (1) (1) - Orange

Evaluation Parameters

Number of folds: 10

Forecast steps: 5

Apply

7133

	RMSE	MAE	MAPE	POCID	R^2	AIC	BIC
ARMA(1,0,0)	125.7	69.5	0.637	63.3	-0.053	84846	84867
ARMA(1,0,0) (in-sample)	104.9	63.8	0.617	43.8	-0.096	85443	85463

Model Evaluation (1) (1) (1) - Orange

Evaluation Parameters

Number of folds: 10

Forecast steps: 5

Apply

1551

	RMSE	MAE	MAPE	POCID	R^2	AIC	BIC
ARMA(1,0,0)	160.2	97.8	0.608	53.1	-0.052	19048	19064
ARMA(1,0,0) (in-sample)	146.0	98.7	0.669	47.7	-0.053	19700	19716

Conclusions and Future Work

Conclusions:

- Demonstrated the effectiveness of the ARIMA model for price forecasting using the Orange data mining platform.
- Captured significant trends and seasonal variations in food prices across various markets in the Philippines.
- Enhanced access to data-driven decision-making for stakeholders in the agricultural sector.
- Emphasized the importance of rigorous model evaluation using metrics (RMSE, MAE, MAPE, R^2, AIC, BIC).

Conclusions and Future Work

Future Work:

- **Integration of Machine Learning:** Combine ARIMA with ML techniques for improved accuracy.
- **Incorporating External Variables:** Include economic indicators or climate data for better predictive power.
- **Real-time Forecasting:** Develop systems for immediate price forecasting insights.
- **User Feedback:** Gather user feedback on the Orange platform for iterative improvements.
- **Alternative Models:** Compare ARIMA with other forecasting models to assess strengths and weaknesses.

Thank you for listening!